



Investigation report

Grounding on Boolarong Bank, Weerona Channel, Kanooka Heads - ATSB final report

Drafted 30/09/2026

INVESTIGATION WORKFLOW SUITE

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Investigation summary

What happened

On 07/07/2026, the 24.4 m aluminium catamaran passenger ferry MV Tarrakin Star was inbound to Kanooka Heads on service TF18, which the winter timetable had moved into darkness. It carried 38 passengers, a master, a deckhand and a kiosk attendant, with the master alone on the bridge. At 18:07:06 in the Weerona Channel, the chart plotter, the passage plan's primary means of position fixing, froze and did not recover. The master continued at about 8 kt, searching by eye for the No. 4 beacon, while the heading increased from 195° to 200° and was then held. The vessel crossed the channel's western edge at about 18:09:00, 114 seconds after the freeze. The echo sounder's shallow alarm sounded at 18:11:48 and both throttles were moved to astern. At 18:12:04 the vessel contacted Boolarong Bank at 6.8 kt, coming to rest 74 m west of the charted centreline and 54 m outside the dredged channel.

The master stopped both engines and did not attempt to refloat on the falling tide. There were no serious injuries: one passenger reported a grazed shin and two were treated at the wharf for distress. All passengers were ashore by 19:26. The starboard hull was split, the No. 2 void flooded slowly, and the starboard propeller, shaft, rudder stock and skeg were damaged. The vessel was refloated at 00:20 on 08/07/2026 and withdrawn from service. The Weerona Channel was closed to commercial traffic for 19 hours.

What the investigation found

The operator had no night passage plan for the Weerona Channel, no hold point if the plotter was lost, and nothing directing the master to stop, which very likely contributed. The winter timetable change likely contributed by moving TF18 into darkness without reopening a route risk assessment that restricted the channel to daylight transits, although a review might have produced a plan no better than the daylight one.

With the plotter frozen, the No. 4 beacon was the only means of fixing available. Mangrove canopy on Weerona Islet screened it from the channel between 0.84 nm and 0.34 nm north of it. The master's search for it over that stretch very likely contributed to the vessel leaving the channel. Clearing of that sightline had been deferred, unfunded, for three works rounds, which very likely contributed. The beacon's solar conversion had halved its nominal range without any operator being told. This likely contributed, although a brighter light would not have been seen where it was screened. The waterway's regulatory framework, with no availability standard and no duty to notify users, likely contributed, with less confidence than the other factors.

Both masters knew of the plotter fault at the change of master, but it was not entered in the defect book, as the verbal handover practice allowed. Both likely contributed, because earlier freezes never reached anyone who could apply an existing firmware fix, although a single entry on the day would not by itself have stopped TF18 sailing. Continuing at about 8 kt after the freeze very likely contributed. The echo sounder alarm, set at 0.9 m under the keel, gave 16 seconds' warning against a 92 m crash-stop distance, which very likely contributed.

The forward saloon lifejacket locker could not be reached from most of the saloon's seats; this did not contribute but very likely increased risk. The winter roster likely increased risk, although there was no indication that fatigue affected the master on the night. Water level and steering or propulsion failure were ruled out. Whether new foreshore floodlighting behind the beacon affected its detection could not be resolved.

What has been done as a result

No safety action has been recorded in response to the safety issues identified.

Safety message

A passage plan that names a primary means of position fixing should say what to do when it is lost; on a narrow channel, stopping is usually safest. Timetable changes that alter a route's conditions, such as moving a transit into darkness, should prompt a review of the existing risk assessment. Intermittent navigation equipment faults should be written down even when they appear to have cleared, so they reach people ashore who can act on them.

The occurrence

All times are local time unless otherwise stated.

The voyage and the change of master

On 07/07/2026 MV Tarrakin Star, a 24.4 m aluminium catamaran passenger ferry certificated for 96 passengers, was operated by Tarrakin Ferries between Muntari Island and the town wharf at Kanooka Heads. At about 13:05 at the town wharf, the relieving master took over from the morning master.

At the handover, the outgoing master said the bridge chart plotter had frozen twice that morning, outside the Weerona Channel, recovering by itself after a minute or two each time. The plotter's event log recorded watchdog resets at 10:42 and 11:17, each with cause code 0x51 (display process not responding). The relieving master recalled replying that the plotter had done the same to him the month before; both masters' accounts agreed. Neither entered the fault in the wheelhouse defect book, which held no entry for 07/07/2026 before the occurrence.

The afternoon's two round trips were described as normal. TF18, the last service of the day, was one of the winter timetable's evening returns, made in darkness every day since 01/06/2026. Sunset was at 17:02 and civil twilight ended at 17:29. The master described clear conditions, a light westerly and flat water. TF18 departed Muntari Island on time at 17:35 with 38 passengers, a master, a deckhand and a kiosk attendant. The master reported the departure to Kanooka Heads Port Control on VHF channel 12, which replied that nothing would affect the vessel and the channel was clear.

Entering the Weerona Channel and loss of the plotter display

At 18:03:44 the master remarked that the No. 1 beacon was coming up. The shore AIS receiver recorded the vessel abeam it at 18:03:58, 2 m east of the charted centreline, at 8.4 kt on a course of 195°. At about 18:04 the master sent the deckhand to collect fares, saying he would call when approaching the wharf, and was thereafter alone on the bridge.

At 18:05:30 the vessel was abeam the No. 2 beacon, 1 m west of the centreline and 0.90 nm north of the No. 4 beacon, at 8.4 kt. The bridge audio recorded the master saying, "Two ... good."

At 18:07:06 the plotter's event log recorded that the display process was not responding; unlike that morning, no reset followed. The master described the chart remaining on screen with the vessel symbol stationary and the display not responding to touch. The bridge audio recorded "Oh, come on" at 18:07:11, sounds consistent with repeated screen taps at 18:07:26, and a partly unintelligible remark ending "frozen again" at 18:07:44, when the vessel was 0.68 nm north of the No. 4 beacon and 3 m west of the centreline.

The master continued at the same speed, looking by eye for the No. 4 starboard-hand beacon on the western side of the channel. He stated he did not consider stopping, regarding the visual transit as the backup. At 18:08:31, 0.47 nm north of the beacon and 12 m west of the centreline, the audio recorded: "Right. Where's four." At 18:09:40 it recorded a sound consistent with a window or door being slid open; the master stated he opened the wheelhouse window in case the glass was the problem, and still could not find the beacon. He described the view ahead as a single glow from the town and the new car park floodlighting, against which the green flash was hard to pick out, but said he could not tell whether the lighting was why he could not see it, only that it was harder to see than it used to be.

The AIS track showed the heading increasing steadily from 195° to 200° over about a minute from 18:07:06 and then being held, with the rate of turn peaking at 2.1° per minute at 18:07:40 and

returning to zero by 18:08:50. The vessel moved west at an average of 12 m per minute, reaching the channel's western edge, 20 m west of the centreline, at about 18:09:00, and was outside the dredged channel thereafter. The engine monitoring recorder showed both engines at 1,780 and 1,784 rpm with no alarm and no throttle movement. Steering hydraulic pressure stayed between 96 and 99 bar, within its normal 90 to 105 bar band, and rudder angle followed helm demand to within 1.5°. The master stated he had eased about five degrees to starboard while searching for the beacon, without deciding to do so.

At 18:10:02 the audio recorded: "There's four. Keep her over." The vessel was 0.25 nm north of the beacon and 39 m west of the centreline. The master recalled the beacon appearing dead ahead rather than on the starboard bow, and understanding at once that he was on the bank side of the channel. A range and visibility check on 18/07/2026 from the inbound track, at the same time of night, found the beacon could not be seen from the channel between 0.84 nm and 0.34 nm north of it.

The grounding

Speed over the ground fell from 8.4 kt to 7.6 kt between 18:07 and 18:11 with no throttle movement. At 18:11:00 the vessel was over Boolarong Bank, 58 m west of the centreline. At 18:11:48, 70 m west and making 7.1 kt, the echo sounder's shallow alarm sounded; it had been set at 0.9 m under the keel since commissioning in 2019. The master said "No no no" at 18:11:50, and at 18:11:52 both throttles were moved together to the astern detent. Astern thrust had not reached full when, at 18:12:04, the vessel contacted Boolarong Bank at 6.8 kt. In those 16 seconds it covered 58 m; its trials crash-stop distance from 8 kt was 92 m.

The audio recorded a heavy impact followed by about 9 seconds of grinding and vibration. The vessel stopped within 11 m, 74 m west of the charted centreline, 54 m west of the channel's western edge and 34 m west of the No. 4 beacon. The bed at the contact point was later surveyed at 0.50 m below chart datum. Furrows consistent with the starboard hull and skeg ran 11 m from first contact on a heading of 197°. The occurrence notification recorded the vessel heeled 4° to starboard, hard aground on a falling tide.

After the grounding

At 18:12:31 the master stopped both engines. He did not attempt to refloat, citing the falling tide, the unknown bottom and the people on board. At 18:12:40 he told passengers over the public address that the vessel had touched bottom and was stopped. At 18:13:04 he broadcast a pan pan to Port Control on VHF channel 16, reporting the vessel aground on Boolarong Bank abeam the No. 4 beacon with 38 passengers and three crew, no injuries, no pollution, stable and upright, and not attempting to refloat, and requested the workboat, which Port Control called out. At 18:13:47 he advised that the tide was falling and the vessel would settle further.

At 18:14:52 the master told passengers to put on lifejackets, and the deckhand issued them from aft forward. In the aft saloon, passengers opened the locker themselves. In the forward saloon, where the locker is behind the kiosk servery counter, three passengers were standing at the counter and the deckhand handed jackets over it. The deckhand reported all passengers in lifejackets at 18:22:40.

At 18:16:20 the deckhand reported about 75 mm of water on the sounding rod in the No. 2 starboard void. The master reported slow ingress to Port Control at 18:17:05 and had the void sounded every ten minutes. The workboat came alongside the port quarter at 18:41. At 19:26:08, 74 minutes after the grounding, the master reported all 38 passengers ashore, with the three crew

remaining aboard. One passenger reported a grazed shin and two were treated at the wharf for distress; there were no serious injuries.

The vessel was refloated on the rising tide at 00:20 on 08/07/2026, with approximately 260 mm of water in the void, and towed to the town wharf. The starboard hull bottom plating was set up over 4.1 m with a 180 mm split in way of the No. 2 void, the starboard propeller blades and shaft were bent, the starboard rudder stock was distorted 3°, and the starboard skeg was torn away. The vessel was withdrawn from service and the Weerona Channel was closed to commercial traffic for 19 hours.

Context

Personnel information

The master

The master had held a certificate of competency as master of vessels less than 35 m since 2012. He had approximately 19 years in commercial marine operations, 14 as a master and 11 with Tarrakin Ferries, all on the Muntari Island route, with approximately 4,100 Weerona Channel transits. His last route check, on 14/02/2026, was satisfactory, and he had no prior occurrence history. He had made 26 of the 31 darkness transits on service TF18 between 01/06/2026 and the occurrence.

The outgoing master

The morning-shift master on 07/07/2026 held the same certificate class, with approximately 15 years in commercial marine operations, 9 as a master, 6 with Tarrakin Ferries and 4 on the route, and approximately 1,400 channel transits. His last route check, on 09/11/2025, was satisfactory. He had made the other 5 darkness transits. On 07/07/2026 he worked 05:40 to 13:05, on day six of a twelve-day tour.

The deckhand

The deckhand had been with the operator for 7 months and held a certificate of safety training and a coxswain grade 2 qualification. This was her first winter season on the route; her vessel familiarisation and drills were completed in 12/2025. On the day she worked the master's duty from 13:10, but not a split duty.

Roster and duty

The route was worked 12 days on, 2 days off. On 07/07/2026 the master was on his ninth consecutive day, on a split duty of 05:40 to 08:40 and 13:10 to 19:20: 9 hours 10 minutes of duty over a 13 hour 40 minute spread, with an unpaid 4 hour 30 minute break at home. Both were within the operator's limits of 10 hours duty and 14 hours spread, and within the state crewing determination.

The operator asks masters to keep a fatigue diary but does not collect it. The master recorded 6 hours 15 minutes of sleep on the night of 06/07. In interview he described himself as tired, which he said was normal for a split duty, but not fighting to stay awake and not slow. No relief master was rostered on the route.

Vessel information

General

MV Tarrakin Star was built in 2011, last slipped in 03/2025, and completed its annual survey on 11/03/2026 with no outstanding conditions. The chart plotter was not a statutory item on a class 1D vessel on this route.

Chart plotter

The plotter was a 12-inch multifunction display fed by a separate GPS antenna unit, running chart software revision 4.11 on firmware issued in 03/2024.

On 21/07/2026 an approved electronics contractor bench tested the unit for 46 hours with the same antenna and cabling; the fault did not recur and no defect was found. The contractor reported that cause code 0x51 was known to the manufacturer and addressed by firmware issued in 11/2024,

which had not been applied. The operator did not hold the relevant bulletin and had no subscription to the manufacturer's bulletins.

The wheelhouse defect book held eleven entries in the 90 days before the occurrence, none concerning the plotter, GPS antenna, echo sounder, steering gear, engines or propulsion controls. The maintenance system held nothing about the plotter in 2026. The outgoing master stated he believed the plotter had frozen three or four times on this vessel in the past year, to him and other masters, and that none had been written up.

Echo sounder

The shallow alarm was set to 0.9 m under the keel, the value entered at commissioning in 2019 and never changed. The vessel's trials crash-stop figure from 8 kt was 92 m.

Steering and propulsion

Recorded data showed both engines within 12 rpm of each other with no alarm or temperature or pressure excursion. Steering hydraulic pressure stayed in its normal band, and rudder angle followed helm demand to within 1.5°.

On 12/07/2026 the steering system, including both rams, tiller arms, hydraulics and autopilot, was removed and function tested. All components were serviceable, with no internal leakage, air or pre-existing defect, and a full lock-to-lock timing test was passed within the specified period. Damage to the starboard shaft, propeller and rudder dated from the contact; the port hull, propulsion and steering were undamaged.

Lifejacket stowage

The vessel carried 120 adult and 20 child lifejackets in four lockers. The forward saloon locker held 40 adult jackets, in the bulkhead behind the kiosk service counter.

The 2011 approved arrangement had an open bench there, leaving the locker face clear. The counter was fitted in a 2019 refit under a minor modification notice, which did not trigger re-survey of the safety equipment arrangement.

A survey on 16/07/2026 found that from 22 of the 34 forward saloon seats the locker could only be reached by passing behind the counter through a 0.6 m gap at its inboard end, the only route with people standing at the counter. The saloon muster placard directed passengers to the locker, which was not visible from the seats. The same counter is fitted to the operator's two other 24 m vessels.

Waterway and aids to navigation

The Weerona Channel

The channel is 40 m wide and dredged to 3.2 m, on a waterway managed by Kanooka Shire Council through the Kanooka Heads Port Authority. The drying Boolarong Bank lies to the west. The bed at the contact point was surveyed at 0.50 m below chart datum, rising to 0.15 m below datum 30 m further west, a gradient of about 1 in 70.

Regulatory framework

The Waterways Aids to Navigation Code WAN-2, section 4.1, requires the responsible authority to maintain an aid 'so that it performs the function for which it was established'. Section 4.2, applying only to aids on a declared shipping channel, requires recorded availability of 'not less than 99.0 per cent measured annually' and prior notice to mariners of changes to character, position or range. Elsewhere, section 4.3 requires only that the authority 'be able to demonstrate that the aid is

inspected at intervals it has determined and recorded'. The Weerona Channel is not a declared shipping channel, so section 4.2 did not apply to its six beacons, and the Authority was not in breach of the code.

Under schedule 2 of the Council–Authority services agreement, the Authority inspects each lit aid by daylight quarterly and confirms operation monthly by observation from the town wharf, 2.1 nm south of the No. 4 beacon. Vegetation beyond 50 m of an aid 'is not within the scope of this agreement', and there is 'no requirement in this agreement to notify waterway users of a defect, a deferral or a change to an aid'.

The No. 4 beacon

The No. 4 beacon is a starboard-hand beacon on the channel's western side, character Fl G 2.5s, lantern 4.2 m above chart datum.

In 11/2025, under the Council's waterways energy programme, it was converted from mains to a solar lantern of 3 nautical miles nominal range, against 6 previously; the character was unchanged. The hydrographic service was notified on 04/02/2026 and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued and no operator was told. The operator's passage plan recorded the beacon at 6 M.

A check from the port workboat on 18/07/2026, on the inbound track at the same time of night, measured luminous range at 2.6 nm. Between 0.84 nm and 0.34 nm north of the beacon, the sightline passed through mangrove canopy on Weerona Islet, 240 m north-west of the beacon, measured at 5.8 m against 2.1 m in the 2019 aerial survey.

The Authority's register held two work orders for this sightline:

- **WO-118** (09/2023) recorded regrowth screening the beacon from the northern approach and recommended trimming to 2 m on a three-yearly cycle at an estimated \$18,500. It was deferred as unfunded in the 2023, 2024 and 2025 works rounds.
- **WO-176** (03/2026) recorded that the sightline had fully closed on the northern approach. A file note recorded that, at 240 m from the beacon, the work fell outside the 50 m scope of schedule 2.4 and competed with boat ramps and pontoons for general waterways funding.

The monthly log recorded the beacon observed and operating every month from 01/2024 to 07/2026, last on 02/07/2026, from the town wharf, which has no line of sight to Weerona Islet. No defect had been recorded and no notice issued about the beacon. The Authority had no record of any operator enquiry about the channel's aids and had never provided the register to Tarrakin Ferries.

Both masters, unprompted, described the beacon as having become harder to see at night over about two years; neither had reported it. The operator's reporting system had a 'navigation hazard' category, unused on this route in three years, and no master interviewed knew of it.

Foreshore lighting

The Kanooka Heads foreshore redevelopment, approved by the Council's planning unit, was commissioned on 22/05/2026, ten days before the winter timetable began. It includes 14 LED floodlights on 12 m columns lighting a boat ramp and car park 300 m east of the town wharf. From the inbound track, each presented about 4,800 candela toward the channel, none shielded to seaward. From 0.5 nm north of the No. 4 beacon, the array lay behind the beacon within 4° of bearing and 1.5° of elevation. The Authority's aids to navigation officer was not consulted, and no assessment against the channel's aids was required or made.

Environmental conditions

Sunset was 17:02 and civil twilight ended 17:29. The waning crescent moon, 2% illuminated, rose at 06:12.

At 18:00 the Kanooka Heads automatic weather station recorded wind 305° at 8 kt, visibility greater than 10 km, few cloud at 3,500 ft, and no rain since 09:00. At 18:45 the port workboat reported a slight sea with wind ripple only.

The predicted tide at 18:12 was 0.94 m above chart datum, falling from 1.66 m high water at 15:20 to 0.30 m low water at 21:34. The Port Authority gauge recorded 0.91 m at 18:12, with a largest departure from prediction of 0.07 m in the 12 hours to 20:00. This gave 1.41 m over the contact point, against a static draught of 1.35 m and calculated squat of 0.18 m at 6.8 kt.

Procedures

Passage planning

Operations Manual OPS-3 requires that the master 'will conduct the passage in accordance with the plan' (3.2); that the transit be on the charted centreline, 'monitored on the chart plotter', with the No. 2 and No. 4 beacons identified visually before passing the No. 1 beacon (3.4); and that in reduced visibility the master reduce speed and consider anchoring or returning to open water (3.6). OPS-3 does not address a lookout or second person on the bridge.

Appendix A, issue 2 (02/2023), the current passage plan, specifies 8 knots, 195° true inbound, the chart plotter as primary position fixing, and the visual transit of the No. 4 beacon and town wharf leading lights as secondary. It closes: 'Night transits are not scheduled on this route.' It gives no guidance on plotter loss and no abort or hold point, and no night passage plan exists. The leading lights lie 2.1 nm beyond the No. 6 beacon and resolve as two lights only inside the channel's southern third.

Route risk assessment and the winter timetable

Route Risk Assessment RRA-07, issue 2 (03/2023), names Boolarong Bank, a 40 m channel against an 8.2 m beam, and a 'single unlit reach between the No. 2 and No. 4 beacons'. It states: 'daylight transits only. A night transit of the Weerona Channel requires the General Manager Operations' approval and a specific passage plan.'

Change record CHG-2026-014 moved TF18 from 16:15 to 17:35 from 01/06/2026. It assessed a 'timetable change only', recorded 'No new hazard identified', and answered 'No' to whether route risk assessment review was required. Approved by the Operations Manager on 22/04/2026, it did not refer to the daylight-only restriction. RRA-07 was not reopened, General Manager Operations' approval was neither sought nor given, and no night passage plan was written. Both masters and the deckhand stated the change was communicated only through the new roster or times.

Defect reporting and handover

OPS-9 requires defects rendering the vessel unseaworthy or affecting a statutory item to be reported immediately (9.1). All others 'shall be entered in the vessel defect book at the end of the shift in which they are found', reviewed weekly by the Operations Manager (9.2). At a change of master, the outgoing master briefs the relieving master on 'the state of the vessel, any defect, and anything else affecting the conduct of the service' (9.3).

Nothing requires the brief to be written, or the relieving master to confirm a defect has been entered. Both masters independently described the practice: faults present at handover are written up; faults that have cleared are spoken about.

Safety analysis

External Influences

Aids to navigation on this waterway carry no availability standard and no duty to tell users anything

The Waterways Aids to Navigation Code WAN-2 sets two different levels of obligation for aids to navigation. Section 4.2 applies where an aid is established on a declared shipping channel. There, the responsible authority 'must record its availability, must achieve an availability of not less than 99.0 per cent measured annually, and must notify a change to the aid's character, position or range by notice to mariners before the change takes effect.' The Weerona Channel is not a declared shipping channel, so section 4.3 applied to all six of its beacons instead. That section requires only that the authority 'be able to demonstrate that the aid is inspected at intervals it has determined and recorded.' It is virtually certain that this framework applied to the channel. The code's text and the channel's status are documentary and were not in dispute.

The services agreement between Kanooka Shire Council and the Kanooka Heads Port Authority gave effect to section 4.3. It set:

- a quarterly daylight inspection
- a monthly confirmation 'by observation from the town wharf'
- a scope for vegetation management of 50 m from an aid.

It stated that there was 'no requirement in this agreement to notify waterway users of a defect, a deferral or a change to an aid.' The Authority was not in breach of either the code or the agreement.

Each of the downstream conditions affecting the No. 4 beacon developed within this framework and was consistent with it:

- **The solar conversion.** In 11/2025 the beacon was converted to solar. This reduced the nominal range from 6 to 3 nautical miles. A notification was lodged with the hydrographic service on 04/02/2026, and the chart correction was still pending on the day of the occurrence. No local notice to mariners was issued, and no operator using the channel was told. Tarrakin Ferries' passage plan appendix still recorded the No. 4 beacon at 6 M.
- **The sightline clearing.** Mangrove regrowth on Weerona Islet, 240 m from the beacon, was identified as screening the beacon from the northern approach in 09/2023. The clearing work was deferred at three successive works rounds. The Authority's file note recorded that the islet lay outside the 50 m scope of the agreement. The work therefore competed for funding with boat ramps and pontoons rather than with aids to navigation.
- **The monthly check.** The beacon was recorded as observed and operating on the monthly check in every month from 01/2024 to 07/2026. That check was made from the town wharf, 2.1 nm south of the beacon. Weerona Islet is not in the line of sight from there, so the screening could not be seen.
- **Information to the operator.** No notice about the beacon was ever issued to a waterway user. The aids to navigation register was never provided to Tarrakin Ferries.

The beacon's solar conversion without notice to operators and the deferred clearing of its sightline were each found to have contributed to the beacon not being visible from the channel over the half mile before the vessel left it. Those two matters are analysed under Organisational Influences. This subsection considers whether the external framework itself influenced them.

The investigation found that the absence of an availability standard and a notification duty on this waterway likely contributed. The argument in favour is as follows. Each mechanism in the code that could have carried the beacon's condition to the people using it stopped at the boundary of this channel:

- a recorded availability figure
- a notice to mariners of a change in range.

In particular, the notification duty at section 4.2 would have required the halving of the nominal range to be notified before it took effect. That duty would have placed the range change in front of the operator. Instead, the operator's passage plan still carried the superseded figure on the night.

There was also evidence against this conclusion, and it was weighed.

- **Nothing prevented action.** Nothing in the code stopped the Authority from notifying operators, trimming the mangroves or measuring the light. The records indicate that these decisions turned on funding and priority rather than on anything the code permitted or required. On that reading, the code's scope was background to the Authority's decisions rather than a cause of them.
- **An availability target would not have detected the problem.** The 99.0 per cent availability target at section 4.2 would not have identified the problem at all. The lantern was lit and working throughout, and the range and visibility check of 18/07/2026 measured its luminous range at 2.6 nm against a nominal 3. The deficiency was that the light could not be seen from the channel between 0.84 nm and 0.34 nm north of the beacon. No availability figure measures that.

The investigation accepted the second point. It did not rest the finding on the availability target. The finding rests instead on the notification duty, which addresses directly the undisclosed change in range.

On the first point, the investigation accepted that the Authority could have acted without being required to. However, it considered that the framework set the conditions under which each of those choices was made:

- it imposed no duty to notify
- it permitted the Authority to set its own inspection arrangement
- it confined vegetation work to a radius that excluded the islet.

On the balance of the evidence, the investigation concluded that the absence of these obligations likely influenced the conditions affecting the No. 4 beacon. This conclusion is held with less confidence than the other contributing factors in this analysis, and it should be read as such rather than as settled. The contrary argument, that the Authority's funding and priority decisions were the operative cause, was not eliminated.

Independent of that balance, the framework's threshold had a practical outcome. A dredged channel about 40 m wide, used at night by a passenger ferry carrying 38 passengers on the night, was marked by beacons that no party was obliged to measure, record or report on to the people navigating by them.

Organisational Influences

Winter timetable moved the last service into darkness without reopening the route risk assessment

The route risk assessment for the Weerona Channel, RRA-07 issue 2 (03/2023), named its hazards as Boolarong Bank to the west, a 'channel width 40 m against a vessel beam of 8.2 m', and a 'single unlit reach between the No. 2 and No. 4 beacons'. It carried an explicit operating restriction: 'daylight transits only. A night transit of the Weerona Channel requires the General Manager Operations' approval and a specific passage plan.' That restriction was current when the winter timetable was prepared.

Change record CHG-2026-014 moved service TF18 from 16:15 to 17:35 from 01/06/2026. It assessed the change as 'timetable change only', recorded 'No new hazard identified', and answered 'No' to whether a route risk assessment review was required. The Operations Manager approved it on 22/04/2026, one level below the person the restriction named. CHG-2026-014 did not refer to the daylight-only restriction, and RRA-07 was not reopened. No night passage plan was written, and the General Manager Operations' approval was neither sought nor given. With the end of civil twilight at 17:29, every TF18 transit from that date was made in darkness, 31 of them before the occurrence. The only reference to darkness in the operator's Weerona Channel passage plan is its closing line: 'Night transits are not scheduled on this route.'

Both documents are current controlled copies. The Operations Manager gave only a verbal account of how he understood the restriction, and this was the only limitation on how firmly the matter could be established. It was therefore extremely likely that the winter timetable moved the last service into darkness without the route risk assessment being reopened.

The change process asked whether anything about the change was new, not whether the existing assessment still held. On that question the answer could only be 'no', because the route, vessel, crew and berths were all unchanged. A review of RRA-07 against a night operation would have confronted the unlit reach between the No. 2 and No. 4 beacons, which the assessment already identified as a hazard. It would also have required a specific passage plan before the first night service ran. Such a review might still have produced a plan that said no more than the daylight one, so the connection was not certain. It was likely that the change contributed to the absence of a night passage plan and of any hold point for loss of the primary means of fixing. That absence, and its effect on the decision to continue at 8 knots after the plotter froze, is analysed under Risk Controls.

Beacon converted to solar, nominal range halved, and no operator was told

In 11/2025 the No. 4 beacon was converted from a mains supply to a solar lantern under the Council's waterways energy programme. The replacement lantern was rated at a nominal 3 nautical miles, against 6 for the lantern it replaced. The character, Fl G 2.5s, was unchanged, so the light gave no outward sign of the change. The Authority lodged a notification with the hydrographic service on 04/02/2026, and the chart correction was still pending on 07/07/2026. No local notice to mariners was issued, no operator using the channel was told, and the aids to navigation register had never been provided to Tarrakin Ferries. The operator's passage plan appendix, issue 2 of 02/2023 and still current, recorded the No. 4 beacon at 6 M. A measurement from the port workboat on 18/07/2026 gave a luminous range of 2.6 nm. The conversion, the lantern rating, the lodgement and the absence of any notice are all documentary, and the range was measured independently. It is virtually certain that the change occurred and was not communicated to the channel's users.

As a result, the master was working to a written plan that described a light more than twice as strong as the one in the beacon. A weaker light is harder to acquire, and the master searched for it for 91 seconds, from 18:08:31 to 18:10:02. However, those 91 seconds fell almost entirely within the stretch where the light was screened by the mangrove canopy, discussed below. A screened light is not made visible by being brighter. For that reason it was likely, rather than any higher, that

the reduced range contributed to the No. 4 beacon not being visible from the channel before the vessel left it. Whether new foreshore lighting behind the beacon also affected the master's ability to pick it up could not be resolved, and is discussed separately.

The absence of any requirement to notify users of a change in range stems from the framework discussed under External Influences.

Clearing of the beacon's sightline deferred unfunded for three works rounds

The Authority's own register documents the screening of the beacon. Work order WO-118 was raised in 09/2023 for mangrove regrowth on Weerona Islet screening the No. 4 beacon from the northern approach, between about half a mile and a quarter mile. It recommended trimming to 2 m on a three-yearly cycle, at an estimated cost of \$18,500. The work was deferred as unfunded at each of the 2023, 2024 and 2025 works rounds. Work order WO-176, raised in 03/2026, recorded that the sightline from the northern approach had by then closed completely. A file note against WO-176 recorded why. The islet lies 240 m from the beacon, outside the 50 m scope of schedule 2.4 of the services agreement. The work therefore competed for funding with boat ramps and pontoons in the general waterways vote, rather than being funded as aids to navigation work.

The canopy was given as 2.1 m in the 2019 aerial survey and measured 5.8 m on 18/07/2026, against a lantern height of 4.2 m above chart datum. The Authority's monthly check was made from the town wharf, 2.1 nm south of the beacon, from where the islet is not in the line of sight. It recorded the beacon as operating in every month from 01/2024 to 07/2026. It is virtually certain that clearing of the sightline was deferred as described.

The check on 18/07/2026 found that the light was not visible at all from the channel between 0.84 nm and 0.34 nm north of the beacon. The vessel was 0.68 nm north when the plotter froze and 0.47 nm north when the master began looking for the beacon. It crossed the channel's western edge at about 0.40 nm, and the master sighted the beacon at 0.25 nm, just inside the point where it re-emerges. The light was not dim or defective over that stretch; it was absent. The master might still have failed to acquire it in that minute and a half had it been unscreened. It was very likely that the deferral contributed to the No. 4 beacon not being visible over the half mile before the vessel left the channel.

Risk Controls

No night passage plan for the channel, and no hold point if the primary means of fixing is lost

The Weerona Channel transit was governed by OPS-3 and its appendix A, issue 2 (02/2023), which was the current issue. Section 3.4 required the transit to be 'made on the charted centreline, monitored on the chart plotter'. The appendix named the 'Primary means of position fixing: chart plotter' and a 'Secondary means: visual transit of the No. 4 beacon and the town wharf leading lights'. Its only reference to darkness was the closing note: 'Night transits are not scheduled on this route.'

There was no night passage plan for the channel. Nothing in the manual or the plan said what to do if the plotter was lost during the transit, and the plan named no abort or hold point. The operator confirmed that no night plan existed, so it is virtually certain that this gap in the risk controls existed. The secondary means was also narrower in practice than it appeared on paper. The town wharf leading lights lie 2.1 nm beyond the No. 6 beacon and do not resolve as two lights until the vessel is inside the channel's southern third. At the point where the plotter froze, the No. 4 beacon was therefore the whole of the secondary means. The only other relevant provision, OPS-

3 3.6, required a reduction in speed and consideration of anchoring 'In reduced visibility'. That condition did not apply on the night.

The display froze at 18:07:06, and the vessel crossed the channel's western edge at about 18:09:00. The master therefore had 114 seconds, on flat water with no other traffic, in which a named hold point would have applied. Both masters interviewed said independently that the right response to losing the plotter was to stop. Both also said that nothing tells a master to do so. The investigation allowed for a limiting argument: a master who did not think of stopping unprompted might not have applied a written rule under the same pressure. Weighing that against the time available and the conditions, it was very likely that the absence of a hold point contributed to the transit continuing at eight knots after the plotter froze. How that plan came to be missing is set out under the winter timetable change in the preceding subsection. The individual action that followed from it is discussed under Individual Actions.

Change-of-master defect brief is verbal, with nothing checking that what is said gets written down

OPS-9 required immediate reporting only of a defect that 'renders the vessel unseaworthy or which affects a statutory item of equipment'. The chart plotter is not a statutory item on a class 1D vessel on this route, although the passage plan names it as the primary means of position fixing. All other defects were to be entered 'at the end of the shift in which they are found'. For a fault that comes and goes, that provides no real trigger. Section 9.3 required the outgoing master to brief the relieving master on 'any defect'. It required nothing of that brief to be written down, and nothing of the relieving master afterwards.

Both masters independently described the same understanding. The defect book is for what is broken at handover, and what has come good is spoken about. The manual bears this out, so it is virtually certain that the arrangement existed. This explains why the plotter fault passed verbally at 13:05 and was not booked. It does not make the omission any less significant, and that individual action is discussed separately.

The investigation considered two readings of whether the arrangement influenced the grounding.

Against a contribution. A defect book entry made at 13:05 would have been reviewed by the Operations Manager weekly, not that afternoon. The vessel would have sailed TF18 either way. Any path from a written entry to a different outcome on the night depended on a same-day decision that nobody in the system was required to make.

For a contribution. The significance lies not in the single entry but in the pattern behind it. The outgoing master believed the plotter had frozen three or four times on this vessel in the past year, to him and to others, and that none of those events had been written up. The maintenance system held no plotter entry of any kind for 2026. The fault code logged that morning is addressed by manufacturer firmware issued in 11/2024. The operator did not hold that bulletin, had no subscription to the manufacturer's bulletins, and had not applied the fix. Because the fault never reached anyone ashore who could act on it, the master met it on the night with no reason to treat it as more than a nuisance.

The investigation accepted the second reading while recognising the force of the first. It was likely that the arrangement contributed to the plotter fault going unrecorded, and through that to the freeze on the night.

Forward saloon lifejacket locker not reachable from most of the saloon it serves

The forward saloon's locker holds 40 adult lifejackets. It is mounted in the bulkhead behind the kiosk server counter. While issuing lifejackets, the deckhand had to ask passengers standing at

the counter to move and then pass behind it through a 0.6 m gap. At 18:22:40 the deckhand reported that 'the locker's behind the counter and there's people standing at it.'

The survey of 16/07/2026 found that the locker face cannot be reached from 22 of the 34 forward saloon seats without passing behind the counter. With people standing at the counter, that gap is the only route. The muster placard directs passengers to the locker, which cannot be seen from their seats. It is virtually certain that this condition existed.

The arrangement approved at build in 2011 had an open bench in that position with the locker face clear. The counter was fitted in a 2019 interior refit carried out under a minor modification notice. That notice did not trigger a re-survey of the safety equipment arrangement. The same counter is fitted to the operator's two other 24 m vessels.

It was unlikely that the locker position influenced the occurrence or its outcome. All 38 passengers were in lifejackets by about 18:22, and all were ashore by 19:26. It was nonetheless very likely to be important. In a fire, a night evacuation into the water, or an event that injured the deckhand, a saloon whose passengers cannot reach their lifejackets without crew help would have to wait rather than equip itself. The underlying weakness is the modification route. An interior refit changed a safety equipment arrangement, and nothing in that route asked whether it had.

Local Conditions

Chart plotter display froze during the channel transit and did not recover

It is very likely that the chart plotter display froze at 18:07:06 and did not recover. Three independent sources agree:

- The plotter's internal event log records the display process not responding at that time, with no reset.
- The bridge audio records the master saying 'Oh, come on' at 18:07:11, then the sound of repeated screen taps at 18:07:26, then '(unintelligible) frozen again' at 18:07:44.
- The master stated that the chart picture stayed on the screen with the vessel symbol stationary, and that the display did not accept touch input.

The log recorded the same cause code, 0x51, at 10:42 and 11:17 that morning. On both of those occasions the unit recovered itself.

The evidence has a limit. The unit was bench tested for 46 hours on 21/07/2026 with the same antenna and cabling. The fault did not recur and no defect was found. The event log is therefore the software's own report of itself, not an independently demonstrated failure. The contractor did identify cause code 0x51 as a fault known to the manufacturer. Firmware issued in 11/2024 addressed it, but that firmware had not been applied to this unit.

It is extremely likely that the freeze influenced what followed. The vessel held the centreline to within 3 m while the plotter worked. It crossed the channel's western edge at about 18:09:00, less than 2 minutes after the freeze. The master began searching for the No. 4 beacon 85 seconds after the freeze. From that point the transit was made without the primary means of position fixing that the passage plan specified.

Two related matters are analysed in other sections:

- The fault was passed on verbally at the change of master and not entered in the defect book. This is analysed under Individual Actions and Risk Controls.
- The decision to continue at eight knots rather than stop is analysed under Individual Actions.

No. 4 beacon not visible from the channel over the half mile before the vessel left it

It is extremely likely that the No. 4 beacon could not be seen from the channel between 0.84 nm and 0.34 nm north of it.

The range and visibility check on 18/07/2026 established this. It was made from the port workboat, on the inbound track and at the same time of night. Over that stretch, the sightline to the beacon passes through the mangrove canopy on Weerona Islet, 240 m north-west of the beacon. The canopy measured 5.8 m high, against a lantern height of 4.2 m. The light is visible beyond 0.84 nm, where the sightline passes the islet's northern edge, and again inside 0.34 nm, where it opens past the southern edge.

This finding is not rated higher because the check was made 11 days after the grounding and the canopy was not surveyed on the night.

The recorded audio is consistent with the measurement:

- The master said 'Right. Where's four' at 18:08:31, when the vessel was 0.47 nm north of the beacon.
- He said 'There's four. Keep her over' at 18:10:02, at 0.25 nm.

The two remarks were 91 seconds apart, and the vessel crossed the channel's western edge between them. The search therefore began inside the screened arc, and the beacon was sighted shortly after the sightline opened. Both masters interviewed raised the beacon without being prompted. Neither had ever reported it.

It is very likely that this condition influenced the vessel leaving the channel. With the plotter frozen, the beacon was the only means of fixing position available. It was not visible during most of the period in which the vessel set west. This assessment is not rated higher because the vessel had already begun to set west before the master started looking.

Over the half mile that mattered, the secondary means of position fixing named in the passage plan was not available to be seen.

The conditions that produced the screening are analysed under Organisational Influences:

- the solar conversion that halved the beacon's nominal range without users being told
- the three consecutive deferrals of the sightline clearing work.

New foreshore floodlighting behind the beacon reduced the master's ability to pick it up

The investigation examined whether new foreshore floodlighting made the beacon harder to detect. It could not resolve the question.

Evidence supporting the proposition

- The master described looking 'into a wall of light', and said the beacon was 'a nothing' against the town and the new car park.
- The outgoing master, independently, said the light was 'washed out altogether from up the top'.
- The survey on 18/07/2026 found 14 unshielded LED floodlights on 12 m columns at the new boat ramp. Each presented about 4,800 candela toward the channel.
- The floodlights were commissioned on 22/05/2026, 10 days before the winter timetable began.

- From 0.5 nm north of the beacon, the array lies behind it within 4 degrees of the same bearing and within 1.5 degrees of the same elevation.
- The array is lit continuously, while the beacon shows a green flash of 0.4 seconds every 2.5 seconds.
- The Port Authority's aids to navigation officer was not consulted on the development, and no assessment against the channel's aids was required or made.

Evidence against the proposition

- The one point at which the alignment was measured, 0.5 nm north, lies inside the arc screened by the mangroves. Over that stretch the beacon would not have been visible against a dark background either.
- The master sighted the light once the sightline had opened, with the floodlights in the same position and no dimmer.
- There is no measurement of what could be seen on the night, only the survey made 11 days later.
- Nothing establishes a detection threshold for a 0.4-second green flash against that background.

Why the matter could not be settled

The alignment is real, and the two masters' accounts are consistent and independent. However, the evidence does not show that the floodlights, rather than the mangroves, prevented the master from seeing the light. While the canopy screens the beacon over the relevant stretch, the question may not be answerable at all.

The investigation assessed it as about as likely as not that the floodlighting reduced the master's ability to detect the beacon. The matter remains unresolved and no finding is drawn from it. This does not remove the case for assessing foreshore lighting against the channel's aids to navigation.

Water level below prediction reduced the depth over Boolarong Bank

It is very unlikely that the water level was lower than predicted. The Port Authority's gauge recorded 0.91 m at 18:12, against a prediction of 0.94 m. The vessel touched because it was 54 m outside the dredged channel, not because of a set-down in the tide.

Day nine of twelve, on a split duty across a thirteen hour forty spread

It is virtually certain that 07/07/2026 was the master's ninth consecutive day. He was working a split duty of 9 hours 10 minutes across a spread of 13 hours 40 minutes, and he recorded 6 hours 15 minutes of sleep the night before. All of these figures were inside the applicable limits.

The investigation found it unlikely that fatigue influenced this occurrence:

- The master's helm work held the centreline to within 3 m.
- He responded to the shallow alarm in 4 seconds, against a fleet figure of 6 to 9 seconds.
- His radio work after the grounding was complete and correct.
- His decision not to refloat was made quickly and was right.

The one decision fatigue could have degraded was the decision to consider stopping. Both masters independently said that no document asks a master to consider it.

It is likely that the roster nonetheless increased risk. It is a standing exposure that recurs on every darkness transit of the winter timetable.

Individual Actions

Transit continued at eight knots after the plotter froze, rather than stopping in the channel

It is virtually certain that the transit continued at eight knots after the plotter froze. The recorded track, the recorded audio and the master's own account agree. The AIS track shows no reduction of speed after the freeze at 18:07:06. The later fall to 7.6 kt by 18:11 is accounted for by drag from the shoaling water, because the engine recorder shows no throttle movement until 18:11:52. The bridge audio records the master's attention moving to the search for the beacon at 18:08:31 ("Right. Where's four"), 85 seconds after the freeze. Nothing on the audio indicates that stopping was considered at any point.

The master's account in interview and in his safety report was consistent with this. He did not consider stopping. He went straight to the visual transit because he regarded it as the backup. He also said he would have expected beforehand that he would stop in such a situation. The passage plan supported his view of the visual transit, because it named the visual transit of the No. 4 beacon as the secondary means of position fixing. It gave no instruction to hold position while that secondary means was being established.

Nothing in the conditions prevented a stop. There was no other traffic in the channel, the water was flat, the wind was offshore at 8 kt, and the vessel's trials figure for a crash stop from 8 kt was 92 m. The vessel crossed the channel's western edge 114 seconds after the freeze. A stop at any point in that period would have left the vessel in the dredged channel rather than on the bank. On that basis it is extremely likely that continuing the transit contributed to the vessel leaving the channel. It is also extremely likely that it contributed to the shoaling not being detected until 16 seconds before contact. Continuing at eight knots with no means of fixing position was what turned the loss of the plotter into a grounding.

There was no lookout on the bridge during this period. The deckhand had been sent to collect fares at 18:04:14. This was the normal practice on the route, and OPS-3 did not address it either way.

This action followed from conditions set out elsewhere in this analysis. The Weerona Channel had no night passage plan and no hold point for loss of the primary means of fixing, which is discussed under Risk Controls. Once the plotter froze, the master's attention turned to a beacon that could not be seen over that part of the channel, which is discussed under Local Conditions. The decision not to stop is therefore best understood as the point at which those conditions reached the vessel, not as a factor standing on its own.

Plotter fault passed on verbally at the change of master and not entered in the defect book

It is virtually certain that the plotter fault was passed on verbally and not entered in the defect book. The defect book, the maintenance system and both masters' accounts establish this. The plotter's event log records two freezes during the morning shift, at 10:42 and 11:17 local, and the unit recovered from both by itself.

At the change of master at about 13:05, the outgoing master told the relieving master about both freezes. The relieving master replied that the plotter had done the same to him the month before. Neither master entered the fault in the wheelhouse defect book or reported it to the Operations Manager. The defect book has no entry of any kind for 07/07/2026 before the occurrence. The two accounts of the handover agree in every particular, and both masters volunteered the omission before they were asked about it.

The significance of the omission lies in where the fault did not go. The maintenance system holds no work order, fault report or correspondence about the plotter at any time in 2026. The operator

held no subscription to the manufacturer's bulletins. As a result, the firmware issued in 11/2024 to address cause code 0x51 had never been applied to the unit. The fault the master met at 18:07 was one the manufacturer had already fixed. A defect entry would have made the repeated freezes visible to the people ashore who procure firmware and withdraw vessels from service.

The influence of the omission is limited, however, and the analysis limits it accordingly. A single entry made at about 13:05 would have been reviewed on the Operations Manager's weekly cycle under OPS-9. It would not by itself have stopped TF18 that evening. For that reason it is likely, rather than certain, that the absence of an entry contributed to the plotter freezing during the transit.

This was an omission that both masters made and neither was required to avoid. OPS-9 did not require the change-of-master brief to be written down. It did not require the relieving master to confirm that a defect mentioned in the brief had been entered. It made an intermittent fault on a non-statutory item due only at the end of the shift in which it was found. The practice, rather than the two masters, is the matter of concern, and it is discussed under Risk Controls.

Engines stopped and no attempt made to refloat on a falling tide

It is virtually certain that the engines were stopped and no attempt was made to refloat. The recorded audio and the engine recorder cover the whole period, and the survey confirmed the hull breach. Both engines were stopped at 18:12:31, 27 seconds after contact, and were not restarted.

In the pan pan broadcast at 18:13:04, the master stated "I am not attempting to refloat" as a decision already made. He had the No. 2 starboard void sounded, reported about 75 mm of ingress to Port Control at 18:17:05, and had the void sounded every ten minutes afterwards. He addressed the passengers 36 seconds after the impact and again at 18:14:52, and all 38 passengers were issued lifejackets. In interview he gave four reasons for his decision: the falling tide, the unknown bottom, the number of people aboard, and the fact that the vessel could not sink while it was on the bank.

The survey later confirmed a 180 mm split in the starboard hull plating in way of that void. Ingress had reached approximately 260 mm by the time the vessel refloated at 00:20. The breach was in the compartment the vessel was resting on. Refloating into 3.2 m of water with the ingress not yet assessed would have put a flooding compartment into deep water with the passengers still aboard. It is very likely that the decision not to refloat reduced the risk arising from the grounding. The master's actions after the impact were appropriate and were taken promptly.

Occurrence Events

Vessel left the dredged channel and tracked onto Boolarong Bank

It is virtually certain that the vessel left the dredged channel and tracked onto Boolarong Bank. It is also virtually certain that this contributed to the grounding. The recorded AIS track shows where the vessel was throughout the transit. The surveyed grounding position and the furrows in the bed corroborate it independently.

The vessel entered the channel at 18:03:58 and stayed within 3 m of the charted centreline until 18:07:06. That was the time at which the chart plotter's event log recorded the display freeze. From then on:

- the heading increased steadily from 195° to 200° and was then held;
- the vessel set west at an average of 12 m per minute;
- it reached the channel's western edge at about 18:09:00;

- it was already outside the dredged channel when the master sighted the No. 4 beacon at 18:10:02;
- it was over the bank from about 18:11:00; and
- it made contact at 18:12:04, 74 m west of the centreline.

The channel is 40 m wide and dredged to 3.2 m. The grounding position was 54 m beyond the channel's western edge and 34 m west of the No. 4 beacon.

The shape of the track shows how the vessel came to be there. The recorded rate of turn peaked at 2.1° per minute at 18:07:40 and had returned to zero by 18:08:50. That pattern is a heading change applied and then held. It is not the gradual displacement produced by current or wind. The wind was offshore at 8 kt, and there was no cross-set in the channel at that state of the tide. The westing therefore came from the heading on which the vessel was steered, and it began at the moment the plotter froze. Had the vessel remained within the dredged channel, it would not have been over the bank. The grounding followed directly, and only, from the vessel being outside the channel.

Other subsections deal with why the vessel was steered and held on that heading, and why the departure from the channel was not detected visually over the following 3 minutes:

- The continuation of the transit at 8 kt after the freeze is analysed under Individual Actions.
- The No. 4 beacon not being visible from the channel over the half mile before the vessel left it is analysed under Local Conditions.
- Whether new foreshore floodlighting behind the beacon reduced the master's ability to pick it up could not be resolved. It is discussed as an unresolved matter, and nothing from it is relied on here.

Shoaling not detected until sixteen seconds before contact

It is virtually certain that the shoaling was not detected until 16 seconds before contact. The echo sounder alarm log, the bridge audio and the recorded speed agree to the second. It is extremely likely that this contributed to the grounding.

The echo sounder's alarm log recorded that the "Shallow alarm set to 0.9 m under the keel. This is the value written into the setup when the unit was commissioned in 2019 and it has not been changed." The alarm sounded once on the voyage, at 18:11:48. The bridge audio records the tone beginning at that second and the master's reaction ("No no no") two seconds later. Both throttles were moved to astern together at 18:11:52, four seconds after the alarm. Contact followed at 18:12:04, when astern thrust was developing but had not reached full.

The response to the alarm was prompt, and the shortfall did not lie there. At 7.1 kt over the ground the vessel covered 58 m in the 16 seconds between alarm and contact. The vessel's own trials figure for a crash stop from 8 kt is 92 m. The alarm therefore gave less warning than the vessel needed to stop.

That warning could not have been extended at the alarm setting in use. The surveyed bank gradient across the contact section was about 1 in 70. At that gradient, the alarm log recorded that the alarm "could not have sounded appreciably earlier at that setting." The vessel had been over the bank from about 18:11:00. Its speed over the ground had fallen from 8.4 kt to 7.6 kt between about 18:07 and 18:11 with no throttle movement, which was consistent with the increasing drag of shoaling water beneath the hulls. Nothing on the bridge audio indicates that this was noticed, and the first indication of shallowing that reached the bridge was the alarm itself.

The echo sounder did exactly what it was set to do. The limitation lay in the combination of the setting and the speed:

- a 0.9 m under-keel setting over a 1 in 70 gradient;
- a transit speed of about 8 kt; and
- the resulting warning, which was shorter than the vessel's stopping distance.

A setting that gave about 40 seconds' warning would have changed the outcome. So would a transit speed at which 16 seconds was sufficient to stop. In either case it is extremely likely the vessel would not have made contact. The speed itself stems from the decision to continue the transit at 8 kt after the plotter froze, which is analysed under Individual Actions.

Steering or propulsion failure caused the vessel to sheer off track

A steering or propulsion failure was examined because a heading change followed by a steady set out of a channel can resemble a jammed rudder or a lost engine. It was found very unlikely, on several grounds:

- Both engines ran within 12 rpm of each other until 18:11:52 with no alarm, excursion or throttle movement.
- Steering hydraulic pressure stayed within its normal band throughout.
- The rudder followed helm demand within 1.5°, and the rate of turn showed a heading applied and held rather than a sheer.
- The steering system was removed and function tested on 12/07/2026 and found fully serviceable, and the damage to the starboard shaft, propeller and rudder dated from the contact.

Findings

From the evidence available, the following findings are made with respect to the grounding of MV Tarrakin Star on Boolarong Bank, Weerona Channel, Kanooka Heads, on 07/07/2026. These findings should not be read as apportioning blame or liability to any particular organisation or individual.

Contributing factors

- It is very likely that the winter timetable moved the last service, TF18, into darkness without reopening the route risk assessment RRA-07 and its daylight-only restriction. It is likely that this contributed to the grounding by leaving the channel without a night passage plan, although a review might have produced a plan that said no more than the daylight one. (Safety Issue)
- It is virtually certain that there was no night passage plan for the Weerona Channel, and no hold point if the primary means of fixing was lost. It is very likely that this contributed, because nothing directed the master to stop when the chart plotter display froze. (Safety Issue)
- The transit continued at about eight knots after the plotter froze, rather than stopping in the channel, and this was virtually certain. It is very likely that this contributed, because a stop at any point in the 114 seconds before the vessel crossed the channel's western edge would have left the vessel in the channel rather than on the bank.
- Shoaling was virtually certainly not detected until sixteen seconds before contact, because the echo sounder's shallow alarm remained at its 2019 commissioning setting of 0.9 m under the keel. This very likely contributed, because the 58 m the vessel covered in that time was less than its 92 m crash-stop distance from 8 kt.
- It is virtually certain that the change-of-master defect brief was verbal, with nothing checking that what was said got written down. This likely contributed, because the intermittent plotter freezes over the preceding year never reached anyone ashore who could apply the firmware fix already issued. However, a single defect book entry on the day would have been reviewed only weekly and would not by itself have stopped TF18 sailing. (Safety Issue)
- The plotter fault was passed on verbally at the change of master and not entered in the defect book, which was virtually certain. It likely contributed by keeping ashore staff unaware of a fault that the manufacturer had already addressed. The same limitation applies: an entry made at about 13:05, on a weekly review cycle, would not by itself have stopped TF18 that evening.
- The chart plotter display very likely froze during the channel transit and did not recover. This conclusion rests on the unit's own event log, corroborated by the bridge audio and the master's account, because the fault did not recur during a 46-hour bench test and no defect was found. It very likely contributed, because the vessel left the centreline within two minutes of the freeze.
- Under WAN-2, aids to navigation on this waterway virtually certainly carried no availability standard and no duty to tell users anything. It is likely that this contributed, because the notification duty in section 4.2 was the one mechanism that would have put the No. 4 beacon's reduced range before the operator. This finding is held with less confidence than the other contributing factors. Nothing in the code prevented the Port Authority from notifying operators, clearing the sightline or measuring the light. An

availability measure would also not have detected a beacon that was lit but could not be seen. (Safety Issue)

- In 11/2025 the No. 4 beacon was converted to solar, its nominal range was halved from 6 to 3 nautical miles, and no operator was told, which was virtually certain. It is likely that this contributed, because the operator's passage plan still described a 6 M light. However, the light was screened entirely over the stretch where the master searched for it, and a brighter light would not have been visible there. (Safety Issue)
- Between 0.84 nm and 0.34 nm north of the beacon, the No. 4 beacon was very likely not visible from the channel over the half mile before the vessel left it, as measured on site 11 days after the occurrence. With the plotter frozen, this very likely contributed to the vessel leaving the channel, although the vessel had already begun to set west before the master started looking.
- The vessel left the dredged channel on an applied and held heading change and tracked onto Boolarong Bank, which was virtually certain. It is virtually certain that this contributed to the grounding, 74 m west of the charted centreline and 54 m outside the channel.
- Clearing of the beacon's sightline through the mangrove canopy on Weerona Islet was deferred unfunded for three works rounds, which was virtually certain. It is very likely that this contributed, because the canopy was left at 5.8 m against a lantern height of 4.2 m, which screened the beacon from the northern approach. (Safety Issue)

Other factors that increased risk

- It is virtually certain that the forward saloon lifejacket locker was not reachable from most of the saloon it serves. From 22 of the 34 seats, it could be reached only by passing behind a kiosk servery counter fitted in a 2019 refit that did not trigger a re-survey. Although this did not contribute to the occurrence, it very likely increased risk, and the same counter is fitted to the operator's two other 24 m vessels. (Safety Issue)
- The master was virtually certainly on day nine of twelve, on a split duty across a 13 hour 40 minute spread. The recorded evidence gave no indication that fatigue affected his performance on the night. However, it is likely that this roster increased risk, because it puts the last darkness transit of the winter timetable at the end of such duties, with no relief master rostered.

Other findings

- The investigation examined and ruled out the possibility that the water level was below prediction and reduced the depth over Boolarong Bank. The Port Authority gauge recorded 0.91 m at 18:12 against a prediction of 0.94 m.
- The investigation examined and ruled out the possibility that a steering or propulsion failure caused the vessel to sheer off track. Both the recorded engine and steering data and the removal and function testing of the steering system on 12/07/2026 showed no defect.
- The master stopped both engines 27 seconds after contact and made no attempt to refloat on a falling tide with the hull holed. This reduced the risk to the 38 passengers.

The investigation also examined whether the new foreshore floodlighting behind the No. 4 beacon reduced the master's ability to pick the beacon up. The matter could not be resolved, and no finding is made. The alignment of the floodlights behind the beacon was real, and both masters described it consistently and independently. However, over the stretch where the master was

looking, the beacon was screened by the mangrove canopy. No measurement exists of what could be seen on the night. It could not be established whether the floodlighting or the canopy was the reason the light could not be seen.

Safety issues and actions

Aids to navigation on this waterway carry no availability standard and no duty to tell users anything

Aids to navigation in the Weerona Channel carried no availability standard and no duty to tell users anything. The Waterways Aids to Navigation Code WAN-2 applies its section 4.2 duties only to aids on a declared shipping channel:

- an availability target of 99.0 per cent
- an obligation to record availability
- an obligation to notify a change of character, position or range by notice to mariners

The Weerona Channel is not a declared shipping channel. Section 4.3 therefore applied, and it requires only that the responsible authority can show it inspects at an interval it has set itself.

The services agreement set that interval as:

- a quarterly daylight inspection
- a monthly observation from the town wharf

It limited vegetation work to 50 m from the aid, and it imposed no duty to tell any user about a defect, a deferral or a change. The Port Authority was not in breach of the code or the agreement.

This left a 40 m wide channel, used by a 96-passenger ferry, with beacons that nobody was required to measure, record or report on. Under this framework the No. 4 beacon's range was halved in the solar conversion without any operator being told. Clearing of its sightline was also deferred for three works rounds, because the screening canopy lay outside the 50 m scope.

The investigation found that this likely contributed to the grounding. That finding is held with less confidence than the other contributing factors. Nothing in the code prevented the Port Authority from notifying operators, clearing the sightline or measuring the light. An availability measure would also not have detected a beacon that was lit but could not be seen.

No safety action has been recorded in relation to this safety issue.

Beacon converted to solar, nominal range halved, and no operator was told

In 11/2025 the No. 4 beacon in the Weerona Channel was converted from a mains supply to a solar lantern under the Council's waterways energy programme. The change halved the light's nominal range from 6 to 3 nautical miles, and no operator using the channel was told. This was virtually certain. The light's character, Fl G 2.5s, did not change, so the light gave no outward sign of the change. A notification was lodged with the hydrographic service on 04/02/2026, but the chart correction was still pending on the day of the occurrence. No local notice to mariners was issued. On 18/07/2026 the light's luminous range was measured at 2.6 nm.

The issue mattered because the master was navigating to a written passage plan that described a light more than twice as strong as the one actually in the beacon. That plan was Tarrakin Ferries' appendix, issue 2 and still current, which recorded the No. 4 beacon at 6 M. It is likely that this contributed to the grounding. The contribution was limited because, over the stretch where the master searched for the light, the beacon was screened entirely, and a brighter light would not have been visible there. The issue arose within a waterway arrangement that set no availability standard for aids to navigation and placed no duty on anyone to tell users about changes to them.

No safety action has been recorded in response to this safety issue.

Clearing of the beacon's sightline deferred unfunded for three works rounds

Clearing of the sightline to the No. 4 beacon through the mangrove canopy on Weerona Islet was virtually certainly deferred, unfunded, at three consecutive works rounds. Work order WO-118 was raised in 09/2023. It recorded that regrowth was screening the beacon from the northern approach between about half a mile and a quarter mile. It recommended trimming the canopy to 2 m on a three-yearly cycle, at an estimated cost of \$18,500.

At the 2023, 2024 and 2025 works rounds, each entry read 'defer, unfunded, refer next round'. A later work order, WO-176, was raised in 03/2026 and recorded the same sightline as 'now fully closed on the northern approach'. The Port Authority's file note against WO-176 recorded that the islet lies 240 m from the beacon, outside the 50 m scope of schedule 2.4. As a result, the work competed for the general waterways vote rather than the aids to navigation vote.

The 2019 aerial survey recorded the canopy at 2.1 m. On 18/07/2026 it measured 5.8 m, against a lantern height of 4.2 m. Over the northern approach, the light was therefore neither dim nor defective but entirely screened from view between 0.84 nm and 0.34 nm north of the beacon. The vessel left the channel within that stretch, and the deferrals very likely contributed to the grounding. The Authority's own register had documented the condition since 2023. Because of how the work was scoped, it was weighed against general waterways works rather than against the upkeep of aids to navigation.

No safety action has been recorded in response to this safety issue.

Winter timetable moved the last service into darkness without reopening the route risk assessment

It was extremely likely that the operator's winter timetable moved the last Weerona Channel service, TF18, into darkness without reopening the route risk assessment RRA-07. That assessment restricted the channel to daylight transits.

RRA-07 issue 2 (03/2023) allowed a night transit only with the approval of the General Manager Operations and a specific passage plan. Change record CHG-2026-014 moved TF18 from 16:15 to 17:35 from 01/06/2026. It assessed the change as a timetable change only, identified no new hazard, and required no route risk assessment review. The Operations Manager approved it. The person named in the restriction never saw the change. No night passage plan was written.

The significance of this issue lies in how the change was assessed. The process asked whether the change was new rather than whether the existing assessment still held. As a result, a 96-passenger vessel was taken through a 40 m channel in darkness 31 times before the occurrence, contrary to a restriction that was still in force.

It is likely that this contributed to the grounding. A review of RRA-07 against night operation would have reached the unlit reach between the No. 2 and No. 4 beacons, which the assessment already named as a hazard. It would also have required a night passage plan before the first service ran. This cannot be stated with more confidence, because a review might have produced a plan that said no more than the daylight one does.

No safety action has been recorded in response to this safety issue.

No night passage plan for the channel, and no hold point if the primary means of fixing is lost

The operator's Weerona Channel passage plan was written for the daylight transit, and there was no night passage plan for the channel. The plan named the chart plotter as the primary means of position fixing. However, neither the operations manual nor the plan said what to do if the plotter was lost during the transit, and the plan named no abort or hold point.

This was the operator's own defence against the sequence that occurred on 07/07/2026, and it had never been written. At night, where the plotter was lost, the secondary means of fixing consisted of the No. 4 beacon alone. This was because the town wharf leading lights do not resolve as two lights until the vessel is inside the channel's southern third. The only provision for slowing or anchoring, OPS-3 3.6, applied in reduced visibility, which was not the case on the night. A hold point would have cost nothing on flat water with no other traffic. It would also have been available throughout the 114 seconds between the freeze and the vessel crossing the channel's western edge. Both masters interviewed independently told the investigation that the right response was to stop, and that nothing in the operator's procedures directed them to do so. The absence of a night plan followed from the winter timetable change, which moved TF18 into darkness without the route risk assessment being reopened.

No safety action has been recorded in response to this safety issue.

Change-of-master defect brief is verbal, with nothing checking that what is said gets written down

Under Operations Manual OPS-9, the brief from an outgoing master to a relieving master was virtually certainly verbal. Nothing required it to be written down, and nothing checked afterwards that a defect mentioned in it had been entered in the defect book. Section 9.1 required immediate reporting only of defects that made the vessel unseaworthy or affected a statutory item. The chart plotter was not a statutory item on this class 1D vessel, although the passage plan named it as the primary means of position fixing. Section 9.2 left other defects to be entered at the end of the shift in which they were found, which gave no clear trigger for a fault that came and went. Both masters independently described the same practice: the book was for what was broken at handover, and what had come good was spoken about. This explains why the plotter fault was not written up, but it does not excuse it.

The issue likely contributed to the grounding. The outgoing master believed the plotter had frozen three or four times on this vessel in the past year, and none of those occasions was written up. The maintenance system held no plotter entry for 2026. As a result, the firmware fix issued in 11/2024 was never applied, and the master met the fault on the night with no reason to treat it as more than a nuisance. There is also a contrary reading. A single entry made on the day would have been reviewed at the end of that week, and TF18 would have sailed either way. On that view, the handover rule is a weakness across the fleet rather than a cause of this grounding.

No safety action has been recorded in response to this issue.

Forward saloon lifejacket locker not reachable from most of the saloon it serves

On MV Tarrakin Star, the forward saloon lifejacket locker could virtually certainly not be reached from most of the saloon it serves. The deckhand raised the matter in her safety report. While issuing lifejackets after the grounding, she had to ask three passengers to move away from the kiosk counter and then pass behind it through a 0.6 m gap to reach the locker.

A survey of the stowage on 16/07/2026 found that the locker face could not be reached from 22 of the 34 forward saloon seats without passing behind the counter. With people standing at the

counter, the inboard gap was the only route. The muster placard directed passengers to a locker they could not see from their seats.

The arrangement approved at build in 2011 had an open bench in that position, with the locker face clear. The counter was fitted during a 2019 interior refit carried out under a minor modification notice. That notice did not trigger a re-survey of the safety equipment arrangement. The same counter is fitted to the operator's two other 24 m vessels.

The issue played no part in the grounding. After the grounding the vessel was upright, stable and not sinking, and all 38 passengers were in lifejackets within about eight minutes. However, it very likely increased risk. In a fire, a capsized risk, a night evacuation into the water, or with a deckhand injured, passengers would have depended on crew help to reach lifejackets they could not reach themselves. The underlying gap was the modification route, which allowed an interior refit to change a safety equipment arrangement without any check that it had done so.

No safety action has been recorded in response to this safety issue.

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